CoDesign Compass

Final Client Delivery Package

Project Information

Item	Details
Project Name	CoDesign Compass
Project Type	Full-Stack Web Application
Purpose	AI-assisted participatory issue analysis and reporting platform
Delivery Date	May 2026
Prepared For	Client Handover
Development Team	Ricky Chen, Jingwen Zhang, Eric Pan, Jiaxin Li, Shuo Li, Aria Fung, Zhimeng Zhu

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1. Project Overview

1.1 Introduction

CoDesign Compass is a full-stack web platform designed to support structured public participation and collaborative issue exploration.

The system enables users to:

- Explore public or organizational issues
- Submit structured “Why” and “How” responses
- Participate through guided workflows
- Generate AI-assisted insights and reports
- Provide anonymous or account-linked feedback

Administrators can:

- Create and publish issues
- Monitor participation trends
- Generate AI summaries
- Export reports and analytics
- Manage users and engagement campaigns

The platform combines modern frontend technologies, backend APIs, database management, and AI-assisted processing into a single integrated system.

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2. System Architecture

2.1 High-Level Architecture

```
Frontend (React)
        ↓
REST API (Spring Boot)
        ↓
PostgreSQL Database
        ↓
AI Processing Module
```

The system follows a layered architecture to ensure maintainability, scalability, and separation of concerns.

2.2 Technology Stack

Layer	Technology
Frontend	React + Vite
UI/Animation	Tailwind CSS, Framer Motion
Backend	Spring Boot (Java)
Database	PostgreSQL
ORM	JPA / Hibernate
AI Integration	Ollama / OpenAI-Compatible APIs
Charts & Analytics	Recharts / ApexCharts
Deployment	Render
Version Control	GitHub

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3. Delivered Features

3.1 User Features

Feature	Description
Issue Participation	Users can access issues through share links
Guided Why Workflow	Structured reasoning questions
Guided How Workflow	Improvement suggestion workflow
Profile Tags	Users can select reusable profile tags
Login & Registration	Optional account creation
Email & Coupon Support	Incentive distribution after participation
Responsive Interface	Mobile-friendly UI/UX

3.2 Administrator Features

Feature	Description
Dashboard Analytics	Metrics and participation trends
Issue Management	Create, edit, disable, and manage issues
AI Summary Generation	Generate AI-based reports
Word Cloud Visualization	Analyze keyword frequency
User Management	View and manage participants
Email Campaigns	Send vouchers and update emails
Data Export	Export reports and raw data

4. User Workflow

4.1 Participation Flow

```
/share/:shareId
    ↓
Welcome Page
    ↓
Profile Page
    ↓
Why Questions
    ↓
How Questions
    ↓
Thank You Page
    ↓
(Optional) Login / Register
```

4.2 Data Flow

```
User opens share link
    ↓
Issue fetched from backend
    ↓
Submission created
    ↓
Profile information saved
    ↓
Why responses saved
    ↓
How responses saved
    ↓
AI processing endpoint triggered
```

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5. Backend & API Overview

5.1 Backend Architecture

```
Controller → Service → Repository → Database
```

5.2 Main Backend Controllers

Controller	Responsibility
IssueController	Issue lifecycle management
WhyResponseController	Why-response submission
HowResponseController	How-response submission
ProfileController	User profiles and tags
AIDataProcessingController	AI data processing and report generation

5.3 Key API Endpoints

Method	Endpoint	Purpose
POST	`/api/issues`	Create issue
GET	`/api/issues`	Retrieve all issues
GET	`/api/share/{shareId}`	Public issue access
POST	`/api/why`	Submit why responses
POST	`/api/how`	Submit how responses
GET	`/api/profile/{submissionId}`	Retrieve profile
POST	`/api/profile/{submissionId}`	Save profile
GET	`/api/ai-data/{shareId}`	Generate AI data

6. AI Processing Module

6.1 AI Features

The platform includes AI-assisted processing for:

- Aggregated summaries
- Insight generation
- Word cloud support
- Data cleaning and normalization
- Structured report generation

6.2 AI Data Processing Logic

The AI module:

- Fetches latest Why/How responses
- Cleans invalid inputs
- Converts responses into structured QA pairs
- Merges responses into analyzable text

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7. Deployment & Hosting

7.1 Selected Deployment Strategy

Selected Production Deployment:

Render Production Environment

The production environment has been fully deployed under the client-owned Render account and production domain.

This includes:

Component	Hosting Platform
Frontend	Render Static Site
Backend API	Render Web Service
Database	Render PostgreSQL
SSL	Included
Domain Name	Custom `.au` domain

7.2 Why This Deployment Was Selected

The selected deployment strategy provides:

- Stable infrastructure
- Predictable monthly cost
- Simplified maintenance
- Easier troubleshooting
- Better long-term scalability

8. Current Production Cost

8.1 Monthly Cost

Service	Cost
Backend Hosting	~$7 AUD/month
PostgreSQL Database	~$0.3 AUD/month
Frontend Hosting	~$3 AUD/month
SSL Certificate	Included
Total	~$13.6 AUD/month

8.2 Annual Cost

Service	Estimated Cost
Hosting	~$163.2 AUD/year
Domain Name	~$15 AUD/year
Total	~$178.2 AUD/year

Billing and subscription management are fully controlled by the client through the client-owned Render account. All the cost with a "~" are based on the current plan in May, 2026.

8.3 Domain

Production domain:

```
codesigncompass.au
```

9. Testing & Quality Assurance

9.1 Testing Scope

The following components were tested:

Area	Coverage
Backend Services	API and business logic
AI Module	AI integration and validation
Functional Testing	User and admin workflows
Integration Testing	Frontend-backend interaction

9.2 Functional Testing Summary

Administrator Features

Feature	Result
Login	☑ Pass
Create Issue	☑ Pass
AI Summary	☑ Pass
Export Reports	☑ Pass
Email System	☑ Pass

User Features

Feature	Result
Guided Question Flow	☑ Pass
"I Don't Know" Flow	☑ Pass
Profile Saving	☑ Pass
Account Registration	☑ Pass

9.3 Reliability & Robustness

Testing confirmed:

- Stable API behavior
- Proper exception handling
- AI validation robustness
- Graceful failure handling
- Consistent data validation

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10. Maintenance & Support

10.1 Running the System Locally

Backend

```
cd backend
./mvnw spring-boot:run
```

Frontend

```
cd frontend
npm install
npm start
```

Docker

```
docker-compose up --build
```

10.2 Environment Variables

Example:

```
REACT_APP_API_BASE_URL=http://localhost:8080/api
```

Production:

```
REACT_APP_API_BASE_URL=https://codesigncompass.au/api
```

10.3 Troubleshooting

Issue	Suggested Fix
Frontend showing old version	Clear browser cache
API unavailable	Check Render deployment
Profile save error	Verify backend configuration
Deployment inconsistency	Verify environment variables

11. Known Limitations

Despite successful testing and deployment, several limitations remain:

Limitation	Description
AI Variability	Real-world AI output may differ
No Large-Scale Load Testing	High concurrency not tested
Partial Mock Data	Some reports may still use mock data
Frontend Automated Testing	Primarily manual testing

12. Future Improvements

Recommended future enhancements include:

- End-to-end automated testing
- Improved AI report generation
- Advanced analytics dashboards
- Load and performance testing
- Enhanced reporting exports
- Improved accessibility support
- More advanced user management

13. Included Documentation

The following documents are included in the delivery package:

Document	Purpose
User Guide	End-user instructions
Admin User Guide	Administrator operations
Testing Report	Quality assurance evidence
Maintenance Manual	Technical maintenance
Deployment Recommendation	Hosting recommendation
Wiki Documentation	Project knowledge base

14. Ownership & Account Transfer

Item	Status
GitHub Repository	Transferred to client-managed account
Render Deployment	Running under client-owned account
Production Billing	Managed by client
Production Domain	`codesigncompass.au`
SSL Certificate	Active
Deployment Access	Client has full administrative access

The client now has full ownership and administrative control over the production environment, deployment services, domain management, and operational billing.

15. Final Recommendation

The current deployment and architecture are considered appropriate for:

- Small-to-medium scale deployment
- Client-facing usage

- Educational and organizational participation workflows
- AI-assisted collaborative analysis

The Render-only deployment strategy is recommended for long-term operation due to its balance between:

- Stability
- Cost efficiency
- Simplicity
- Maintainability
- Scalability

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16. Conclusion

The CoDesign Compass platform has been successfully developed, tested, and prepared for deployment.

The final system includes:

- Full-stack web infrastructure
- AI-assisted analysis workflows
- User and administrator interfaces
- Reporting and export capabilities
- Deployment and maintenance documentation

The system is considered stable and suitable for client delivery and future expansion.

The production deployment, billing setup, and operational ownership have been successfully transferred to the client environment.

Appendix

Production Website

```
https://codesigncompass.au
```

Production API

```
https://codesign-project.onrender.com/api
```

Production Domain

```
codesigncompass.au
```

Repository & Source Control

```
GitHub Repository (client-owned & team managed)
```